

Federated Simulation Results

MLP vs. CNN vs. GRU, fixed vs. randomized attack severity — all runs from today

Bhuvan Krishna (Tony) · FL-IDS Project · federated_simulation.py, one full day of OpenSky data (2019-01-07, 24 hourly files)

Executive Summary

Six FedAvg simulations were run today across three model architectures (MLP, CNN, GRU) and two dataset versions (fixed-severity attack injection, and a randomized-severity version built to make attacks harder to detect). All runs partition data by region_cell (simulating independent ADS-B ground receivers) and train via standard FedAvg, never sharing raw data between clients.

Three findings matter most: (1) CNN outperforms MLP consistently on both datasets, a real and robust result, not an artifact of one easy run. (2) The randomized-severity fix worked exactly as intended -- it made gps_spoof meaningfully harder to detect (13-14 point recall drop) while leaving the untouched ghost_aircraft attack unaffected, confirming the severity randomization is doing real work. (3) replay detection is stuck around 40% recall across every single run regardless of architecture, dataset, or GRU tuning -- and GRU, specifically built to fix this via sequence context, has not yet closed that gap. A promising mechanism for why was found via a synthetic test (see Section 5) but has not yet been validated on real data.

GRU also introduced a new, structural failure: ghost_aircraft becomes completely undetectable (zero surviving test examples) in both GRU configurations. This is not a tuning problem -- ghost_aircraft is defined by having almost no message history (track_message_count ~ 1), which is fundamentally incompatible with a model that requires several consecutive messages to form one training example.

1. Methodology Recap

- Clients = region_cell partitions (geographic grid cells), standing in for independent ADS-B ground receiver stations.
- FedAvg: each round, participating clients train locally on their own data, then send weight updates (not raw data) to be averaged into a new global model.
- Fixed-severity dataset: every instance of a given attack type uses one fixed magnitude (e.g. gps_spoof always shifts position by the same range).
- Randomized-severity (v2) dataset: attack magnitude is randomized per row into subtle/moderate/severe tiers -- built specifically because the fixed-severity version let models get away with only ever learning to catch obvious corruption.
- GRU requires real chronological sequences (a window of an aircraft's last N messages), not independent rows -- this needed loading additional timestamp columns and a fundamentally different data-preparation path than MLP/CNN.

2. Full Results Table

Metric	MLP (old)	CNN (old)	MLP (v2)	CNN (v2)	GRU (w=10,15rd)	GRU (w=5,30rd)
Global accuracy	76.87%	79.10%	73.88%	76.63%	61.79%	74.53%
Clients built	122	122	121	121	106	115
benign recall	62.14%	66.01%	59.59%	64.46%	60.03%	73.46%
flooding recall	78.93%	84.21%	79.58%	86.12%	62.25%	83.36%
ghost_aircraft recall	99.70%	99.62%	99.80%	99.78%	N/A	N/A
gps_spoof recall	92.81%	94.18%	78.40%	80.91%	80.41%	78.89%
replay recall	42.01%	43.37%	40.71%	41.25%	31.25%	39.71%
velocity_spoof recall	93.08%	92.53%	91.22%	90.58%	72.08%	85.69%

Metric	MLP (old)	CNN (old)	MLP (v2)	CNN (v2)	GRU (w=10,15rd)	GRU (w=5,30rd)
Model size	12.4 KB	16.1 KB	12.4 KB	16.1 KB	18.4 KB	18.4 KB
Total communication	46.47 MB	60.52 MB	46.09 MB	60.03 MB	59.91 MB	130.00 MB

3. Charts

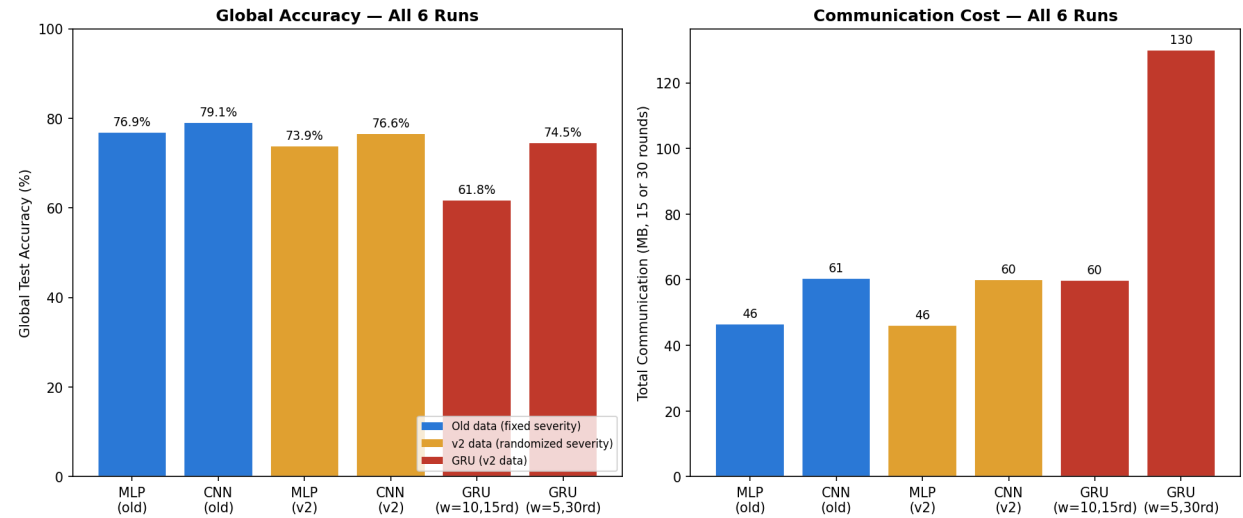


Figure 1. Global accuracy and total communication cost across all 6 runs.

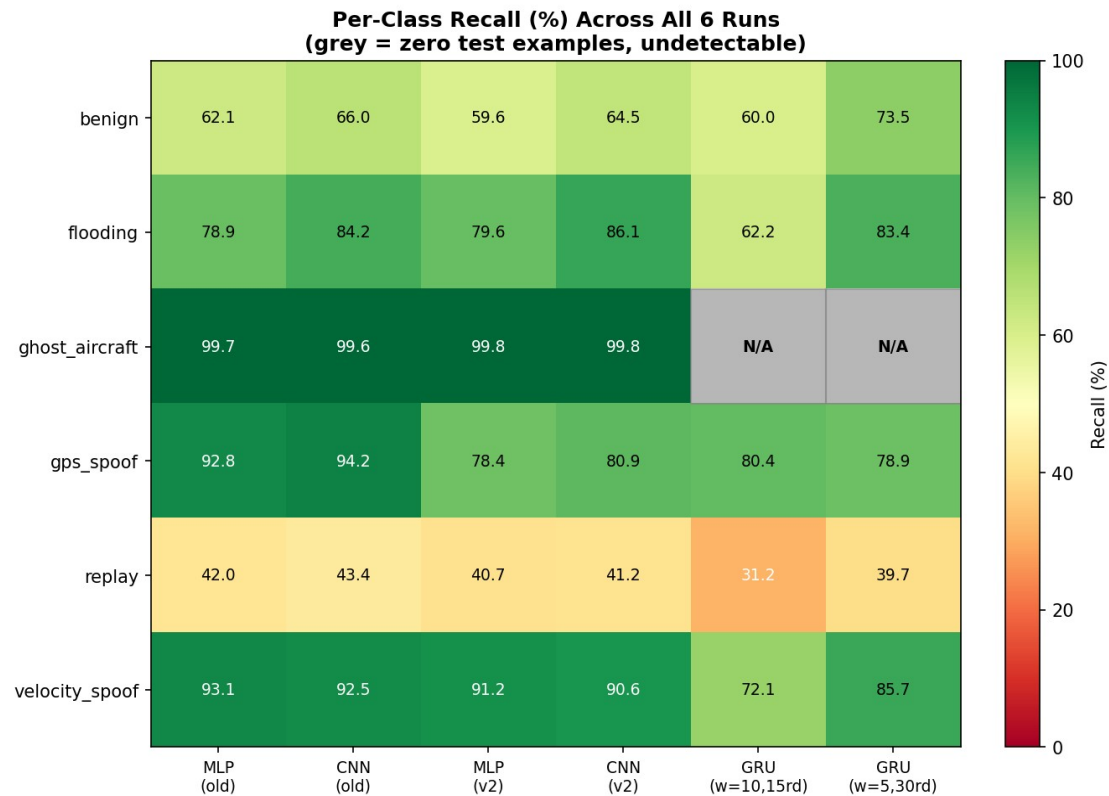


Figure 2. Per-class recall across all 6 runs. Note replay's persistently orange/yellow row and ghost_aircraft's grey (undetectable) cells for both GRU runs.

4. Key Findings

4.1 CNN beats MLP consistently, on both datasets

CNN improves over MLP by a similar margin on old data (+2.23 pts) and v2 data (+2.75 pts) -- this consistency across two meaningfully different datasets is what makes it a real result rather than a fluke of one easy run. The cost is a proportional ~30% increase in communication per round.

4.2 Severity randomization is working exactly as designed

gps_spoof recall dropped 14.41 points (MLP) and 13.27 points (CNN) between old and v2 data -- by far the largest change anywhere in the results, and isolated almost entirely to the one attack type whose severity was deliberately widened into a subtle-to-severe range. ghost_aircraft, untouched by the severity change, stayed flat across both datasets for both models. This is the clean confirmation the randomization fix is doing real work, not just adding noise.

4.3 replay is the one attack no configuration has solved

40-43% recall across MLP and CNN on both datasets. GRU was built specifically to address this via sequence context, but has not beaten either simpler model yet: 31.25% at window=10, improving to 39.71% at window=5 -- back in the same range as MLP/CNN, not ahead of it. Given GRU costs more than double the communication budget to reach that number, its case is currently negative on replay specifically, even though it helped elsewhere (see 4.4).

4.4 GRU's window size mattered enormously -- and revealed a structural blind spot

	window=10, 15 rounds	window=5, 30 rounds
Aircraft dropped (insufficient history)	88.9%	83.6%
Global accuracy	61.79%	74.53%
ghost_aircraft recall	undetectable	undetectable

Shrinking the window from 10 to 5 messages recovered accuracy dramatically (+12.7 pts) by letting far more short-lived aircraft survive into training. But ghost_aircraft stayed completely undetectable at BOTH window sizes -- this is not a tuning problem. A ghost_aircraft is, by design, a fabricated ID that appears almost exactly once (track_message_count ~ 1). No window size can build a multi-message sequence from an aircraft that structurally only ever sends one message. This is a genuine, citable architectural tradeoff: sequence-based detection is structurally blind to identity-scarcity attacks, independent of hyperparameter tuning.

5. Open Lead: Why Does replay Stay Stuck? (Not Yet Validated on Real Data)

A mechanism was identified and confirmed on synthetic data, but NOT yet tested against the real dataset -- flagged clearly as a hypothesis pending validation, not a proven result.

The replay attack manipulates the 'time' field directly (shifting it backward). GRU sequences are currently built by sorting each aircraft's messages by that same 'time' field. A synthetic test showed this can be actively self-defeating: a replayed row's fake, shifted timestamp can sort it AWAY from the final position in its window -- and since the window's label comes from whichever row ends up last, the replayed row can stop being the training target entirely, even though it's still present in the data.

Fix implemented and ready to test: --seq-sort-col lastposupdate sorts sequences by a timestamp field replay does NOT manipulate, keeping each row in its TRUE chronological position. On synthetic data, this correctly kept the attack row in its real terminal position instead of letting it sort away. Not yet run on the real day-1 dataset -- this is the immediate next experiment, not a completed one.

6. Recommendations / Next Steps

- Run GRU with `--seq-sort-col lastposupdate` on the real v2 dataset and compare replay recall directly against the 39.71% (window=5) baseline -- the single most informative next experiment given everything above.
- If lastposupdate meaningfully improves replay, rerun the full window-size sweep with it as the default sort column, since window=5's benefit was measured under the (possibly self-defeating) time-sort assumption.
- Report ghost_aircraft's GRU blind spot as a named architectural finding, not a bug -- it's a legitimate, interesting result about the tradeoff between sequence models and identity-scarcity attacks.
- Given GRU's communication cost roughly matches or exceeds CNN's benefit-per-MB so far, the report's honest framing should be 'GRU trades away ghost_aircraft detection for a currently-unproven chance at fixing replay' rather than presenting it as a strict upgrade.

Limitations

- All 6 runs use one full day of data (2019-01-07) -- not yet validated across multiple days/seasons.
- GRU sequences are built independently per region_cell; a flight crossing a region boundary has its sequence cut off there rather than continuing.
- The lastposupdate hypothesis (Section 5) is supported by a synthetic proof-of-mechanism only -- real-data validation is the immediate next step, not yet completed.
- No centralized-training baseline (same model, pooled data, no FedAvg) has been run yet -- the federated-vs-centralized accuracy gap, standard evidence for the cost of privacy-preserving training, is not currently measured.